

FREDRIK PETTIT

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Education

University of British Columbia

Expected Graduation: May 2028

B.Sc. Computer Science (Artificial Intelligence) | GPA: 3.8

Vancouver, BC

Work Experience

AEM

Apr 2026 – Present

AI Engineer Intern

Langford, BC

- Built a Prefect-scheduled monthly budget classification agent on LangGraph, canonicalizing vendor names and correcting GL codes across **1000+** transactions through rule matching and Claude API reasoning, cutting **6+ hours** of labor to **5 minutes**
- Designed a self-improving categorization system that validates Claude outputs with Pydantic and tracks confidence over time in LangSmith, caching high confidence results as reusable rules and flagging low confidence rows for finance review
- Shipped an evaluation harness that checks each run's categorizations against prior budget-vs-actuals records completed by hand by the finance team, reporting per-vendor accuracy in LangSmith to catch mistakes before the monthly results go out

ELLUSTROS

Jan 2026 – Apr 2026

Software Engineering Intern

The Blue Mountains, ON

- Developed and deployed a FastAPI REST service exposing endpoints for a recommendation engine consumed by a React Native frontend, applying layered filtering and weighted scoring across 15 ergonomic attributes to rank **100+** office chairs
- Cut recurring validation defects by **30%** across the TRÖS Bespoke app by implementing 40+ unit and regression tests in Jest and React Testing Library covering user sign-in flows and AI measurement capture while working in an Agile team

UBC Department of Computer Science

Aug 2025 – Present

Teaching Assistant | CPSC 310, CPSC 110

Vancouver, BC

- Supported **900+** students in UBC's largest computer science course, achieving a 4.9/5 average student evaluation score
- Built 250+ randomized HTML/JavaScript questions and Python autograders for CPSC 310 - Intro to Software Engineering

Projects

YouTube Shorts AI Optimizer | *Python, FastAPI, AWS (Lambda, DynamoDB), OpenCV, XGBoost*

- Developed an end-to-end video optimization pipeline automating thumbnail selection, title, and description generation for YouTube Shorts, supporting a personal channel to grow to **38,000+** subscribers, **45,000,000+** views, and **1,800,000+** likes
- Trained four XGBoost regressors on **3,000+** videos predicting engagement rate to score thumbnail, title, description, and hashtag candidates at 0.043 MAE, thumbnails via OpenCV and AWS Rekognition, text metadata via sentence-transformers
- Orchestrated the pipeline as an asynchronous AWS Step Functions workflow, fanning video-analysis and inference jobs across Lambda functions via SQS queues, then calling Gemini 1.5 Pro for frame selection and Claude API for text generation

Inventory & Sales Automation Platform | *Spring Boot, React, TypeScript, PostgreSQL, Python, FastAPI*

- Built and operate custom infrastructure for a personal cross-platform retail business on Grailed, Depop, and eBay, logging 60+ international orders and **\$6K+** in transactions through Spring Boot API, React/TypeScript frontend, and PostgreSQL
- Automated order logging by parsing Gmail sale confirmations into the tracker, and built a Python/FastAPI Playwright service that reposts 15 listings weekly on a 30-day bump-then-reduce price cycle to maximize buyer visibility
- Improved query performance 9× across 10+ aggregated reporting views by adding targeted indexes, powering the dashboards of a polyglot backend that pairs a Spring Boot cashflow API with a Python/FastAPI Playwright sidecar over JSON/HTTP

Chelsea FC Play Predictor | *Python, pandas, scipy, PostgreSQL, Streamlit, scikit-learn*

- Engineered a system ingesting Chelsea match data from FBref and Understat into PostgreSQL with automated feature computation and matchweek versioning, feeding 7 statistical and ML models producing probability distributions
- Engineered an EV optimizer scoring candidate picks against the app's partial-credit payoff function, validated through backtesting and a £100 competition prize; deployed as a Streamlit dashboard with picks, confidence, and EV breakdowns

Technical Skills

Languages: Java, Python, C++, C#, SQL, R, JavaScript/TypeScript

Frameworks & Libraries: Spring Boot, React, React Native, Expo, FastAPI, JUnit, scikit-learn, XGBoost, LightGBM, pandas, NumPy, scipy, OpenCV, Streamlit, BeautifulSoup, matplotlib, PyQt5, PyTorch, TensorFlow

Cloud & DevOps: AWS (Lambda, Step Functions, SQS, Rekognition), Docker, Gradle, Git

Databases & Tools: PostgreSQL, MySQL, SQLite, DynamoDB, Firebase (Firestore & Storage), Jupyter Notebook